Customer Shopping Behavior Analysis

Data Analysis & Business Intelligence Project

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Role: Data Analyst Project
Tools Used: Python, SQL, Power BI, Pandas, Matplotlib, Seaborn

1. Project Overview

This project analyzes customer shopping behavior to understand purchasing patterns, customer demographics, and product performance. The goal is to generate meaningful insights that can help businesses improve marketing strategies, customer retention, and product planning.

Using Python for data cleaning and exploratory analysis, SQL for structured data querying, and Power BI for visualization, the project follows a complete data analytics workflow from raw data processing to interactive dashboard reporting.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
    - Customer demographics (Age, Gender, Location, Subscription Status)
    - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
    - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Data Processing and Preparation

Before performing analysis, the dataset was cleaned and prepared using Python. Several preprocessing steps were performed to ensure data quality and consistency.

Key data preparation steps include:

- Loading raw data using Pandas
- Standardizing column names
- Handling missing values
- Converting data types
- Creating new features such as age groups
- Creating customer segmentation based on purchase behavior

A Python automation pipeline was developed to perform these steps efficiently and store the cleaned data into a MySQL database.

```
2026-03-11 14:18:14,486 | INFO | ========================================
2026-03-11 14:18:14,486 | INFO | DATA PIPELINE STARTED
2026-03-11 14:18:14,486 | INFO | ========================================
2026-03-11 14:18:14,486 | INFO | Loading Raw Data
2026-03-11 14:18:14,515 | INFO | Data loaded successfully | Shape: (3900, 18)
2026-03-11 14:18:14,515 | INFO | ========================================
2026-03-11 14:18:14,515 | INFO | Initial Data Preview
2026-03-11 14:18:14,515 | INFO | ========================================
2026-03-11 14:18:14,527 | INFO | Initial Preview:    Customer ID  Age Gender Item Purchased  ... Promo Code Used  Previous Purchases Payment Method Frequency of
Purchases
0           1   55   Male        Blouse  ...            Yes                 14         Venmo           Fortnightly
1           2   19   Male       Sweater  ...            Yes                  2          Cash           Fortnightly
2           3   50   Male         Jeans  ...            Yes                 23   Credit Card                Weekly
3           4   21   Male       Sandals  ...            Yes                 49        PayPal                Weekly
4           5   45   Male        Blouse  ...            Yes                 31        PayPal              Annually
5           6   46   Male      Sneakers  ...            Yes                 14         Venmo                Weekly
6           7   63   Male         Shirt  ...            Yes                 49          Cash             Quarterly
7           8   27   Male        Shorts  ...            Yes                 19   Credit Card                Weekly
8           9   26   Male          Coat  ...            Yes                  8         Venmo              Annually
9          10   57   Male       Handbag  ...            Yes                  4          Cash             Quarterly

[10 rows x 18 columns]
2026-03-11 14:18:14,528 | INFO | Data Cleaning Started
2026-03-11 14:18:14,528 | INFO | Column cleaning completed
2026-03-11 14:18:14,529 | INFO | Before: ['Customer ID', 'Age', 'Gender', 'Item Purchased', 'Category', 'Purchase Amount (USD)', 'Location', 'Size', 'Color',
'Season', 'Review Rating', 'Subscription Status', 'Shipping Type', 'Discount Applied', 'Promo Code Used', 'Previous Purchases', 'Payment Method', 'Frequency of
```

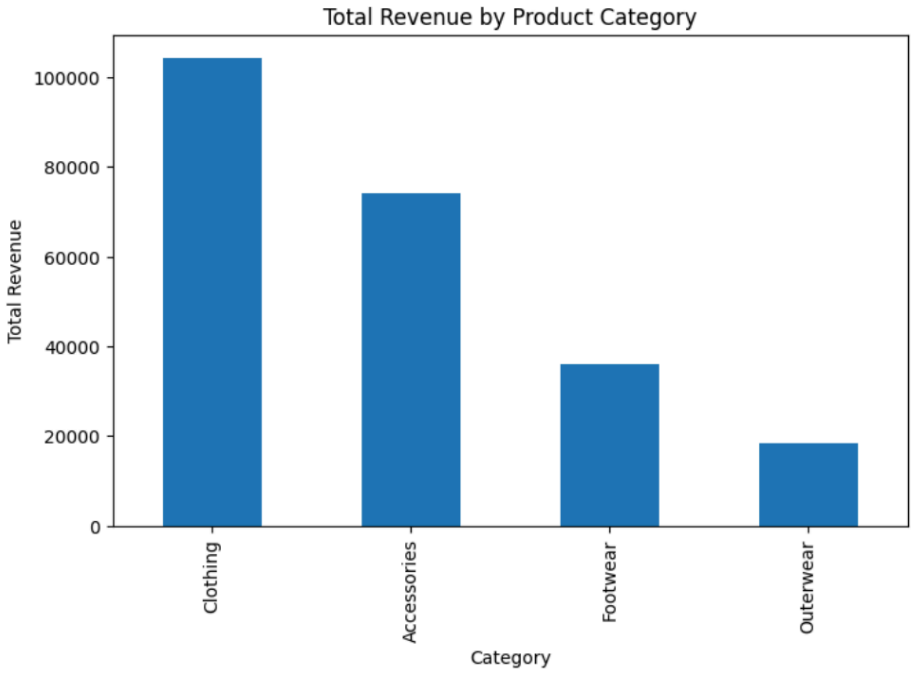
4. Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was performed using Python to understand patterns and relationships within the dataset. Visualization libraries such as Matplotlib and Seaborn were used to generate charts and graphs for better understanding of the data.

The following analyses were performed:

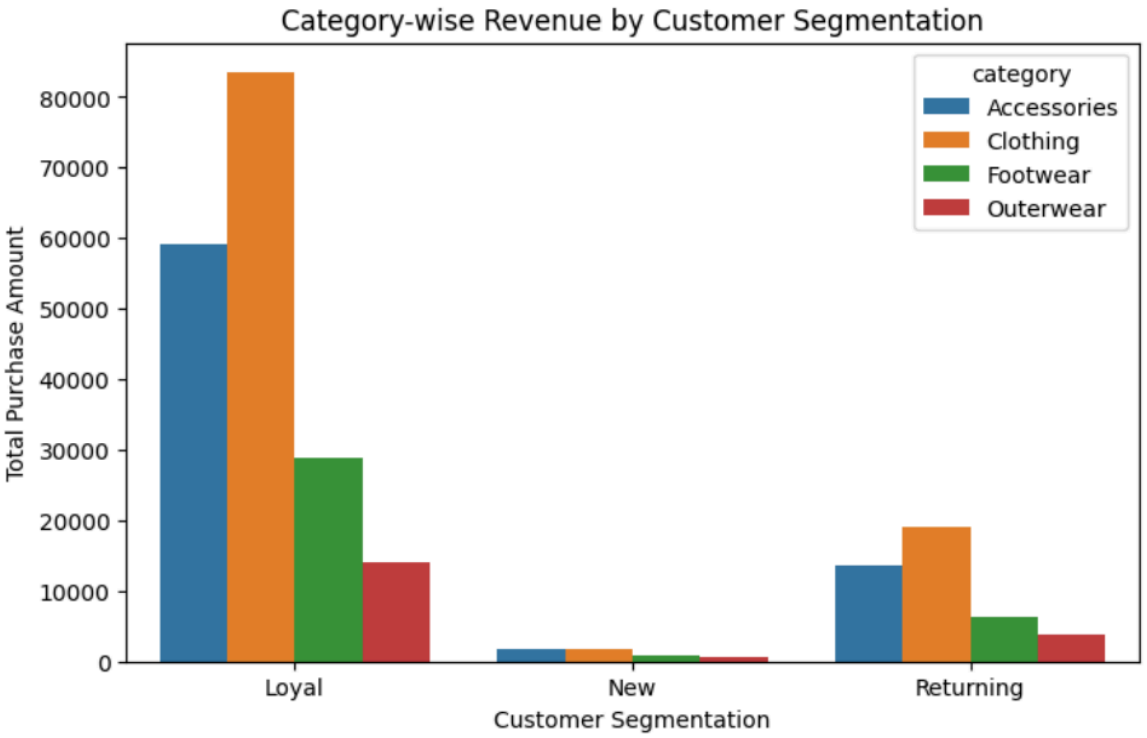
Revenue by Product Category

This analysis identifies which product categories contribute the most to overall sales revenue.



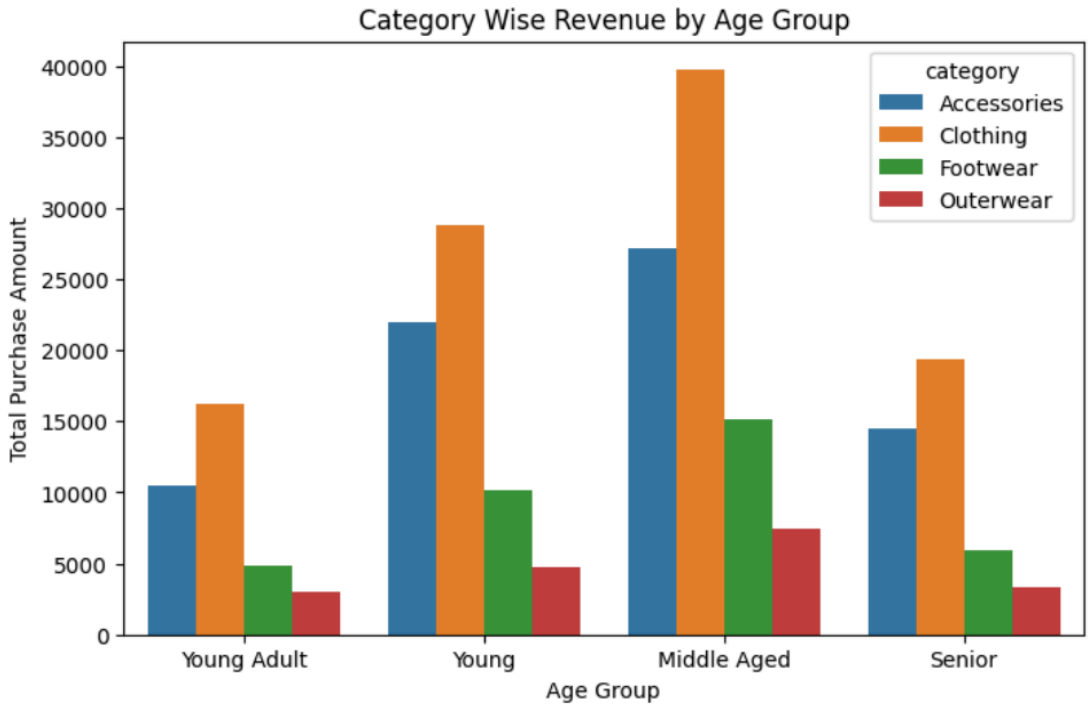
Customer Segmentation Analysis

Customers were segmented into groups based on their purchase history. This helps identify high-value customers and analyze their contribution to total revenue.



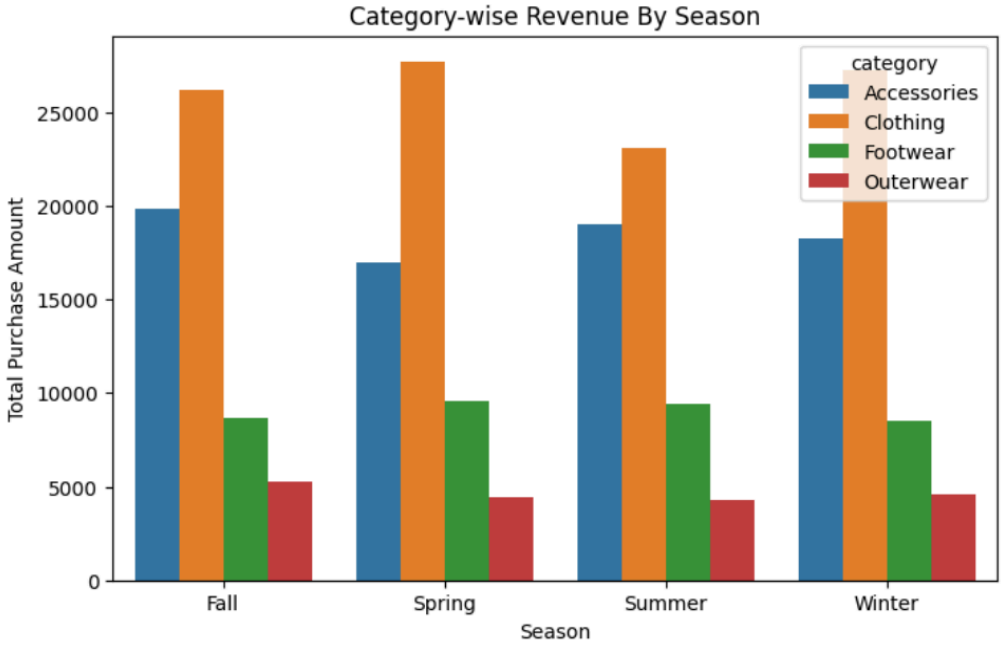
Purchase Behavior by Age Group

Customers were grouped into age categories to understand how spending patterns differ across age groups.



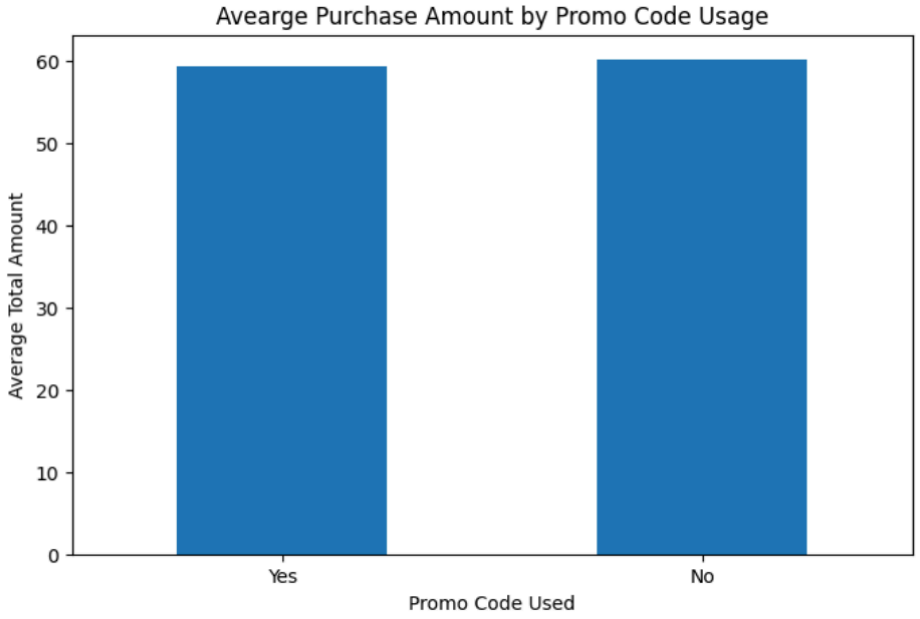
Seasonal Sales Analysis

Sales trends were analyzed across different seasons to identify peak purchasing periods.



Promotion and Discount Impact

The analysis examined whether discounts and promo codes influence customer spending behavior.



5. Data Analysis using SQL

After cleaning the dataset and storing it in MySQL, SQL queries were used to perform additional analysis and extract business insights.

1. **Revenue by Gender-** Compared total revenue generated by male vs female Customer.

[10]:

	gender	customer_count
0	Female	1248
1	Male	2652

2. High-Spending Discount Users - Identified customers who used discounts but still spent above the average purchase amount.

	customer_id	total_revenue
0	2	64.0
1	3	73.0
2	4	90.0
3	7	85.0
4	9	97.0
...	...	...
834	1667	64.0
835	1671	73.0
836	1673	73.0
837	1674	62.0
838	1676	90.0

839 rows × 2 columns

3. Top 5 Products by Rating - Found products with the highest average review rating.

	item_purchased	avg_rating
0	Gloves	3.86
1	Sandals	3.84
2	Boots	3.82
3	Hat	3.80
4	Handbag	3.78

4. Shipping Type Comparison - Compared average purchase amounts between Standard and Express shipping.

	shipping_type	avg_purchase_amount
0	Express	60.48
1	Standard	58.55

5. Subscribers vs Non-Subscribers - Compared average spend and total revenue across subscription status.

	subscription_status	total_revenue	avg_revenue
0	No	170436.0	59.8651
1	Yes	62701.0	59.5451

6. Discount-Dependent Product - Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_percentage
0	Sneakers	50.05
1	Hat	49.02
2	Sweater	48.72
3	Pants	48.23
4	Coat	48.19

7. Customer Segmentation - Classified customers into New, Returning , and Loyal segments based on purchase history.

	customer_segmentation	number_of_customer
0	Loyal	3116
1	New	83
2	Returning	701

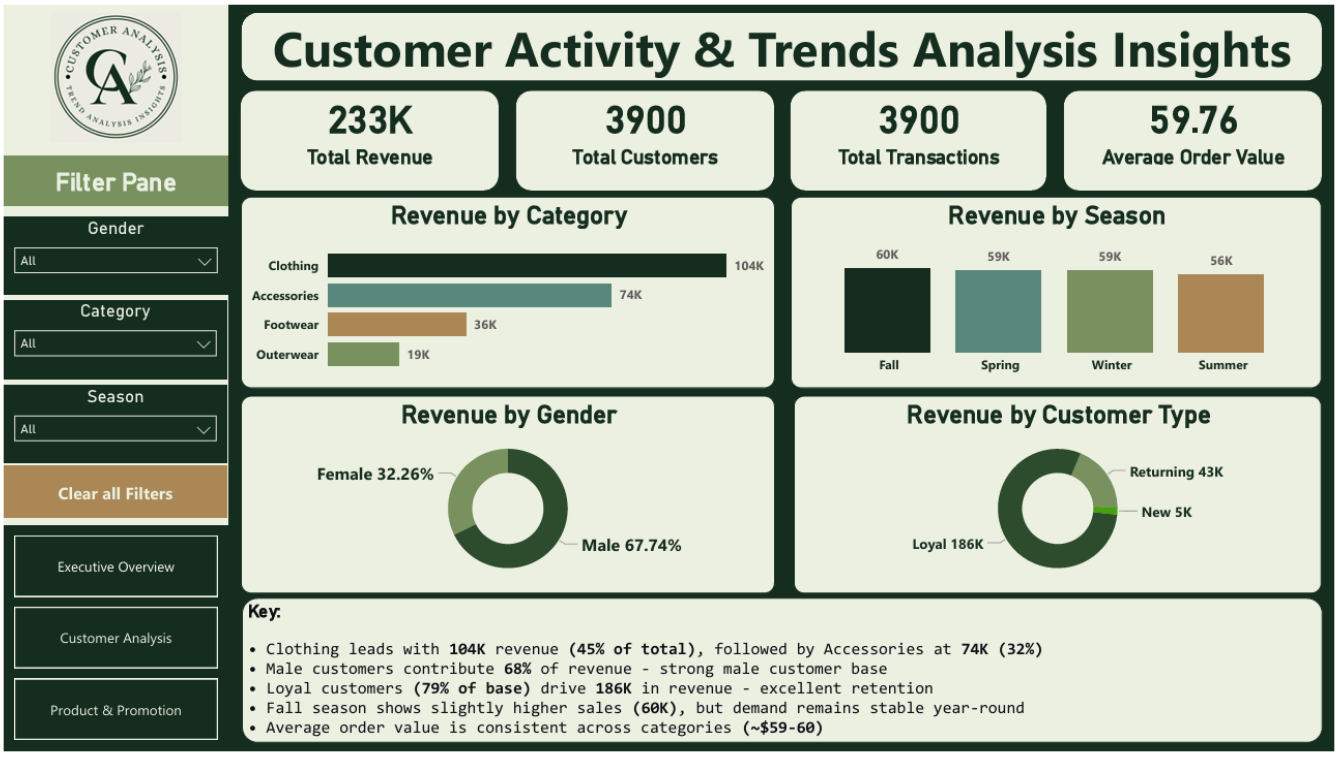
6. Dashboard Development (Power BI)

A multi-page Power BI dashboard was developed to visualize the insights generated from the analysis. The dashboard presents key metrics and trends in a clear and interactive format.

The dashboard includes the following sections:

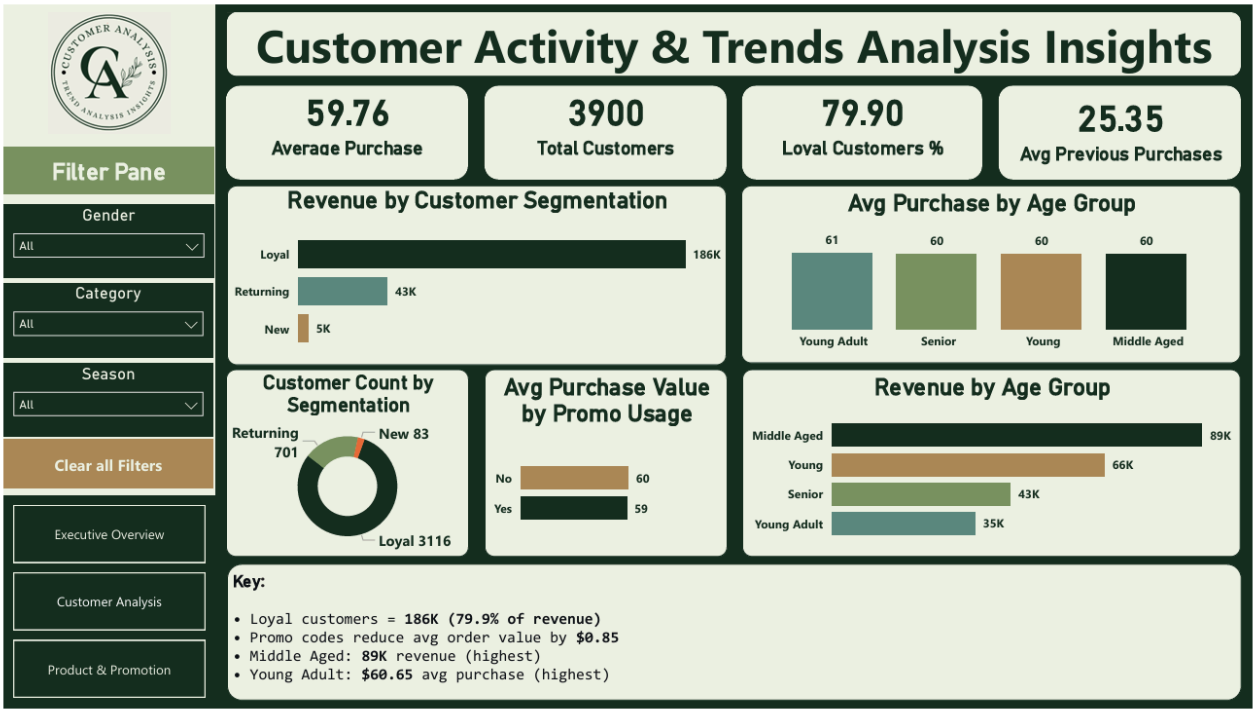
Executive Overview

- Total Revenue
- Total Customers
- Total Transactions
- Revenue by Category
- Revenue by Season



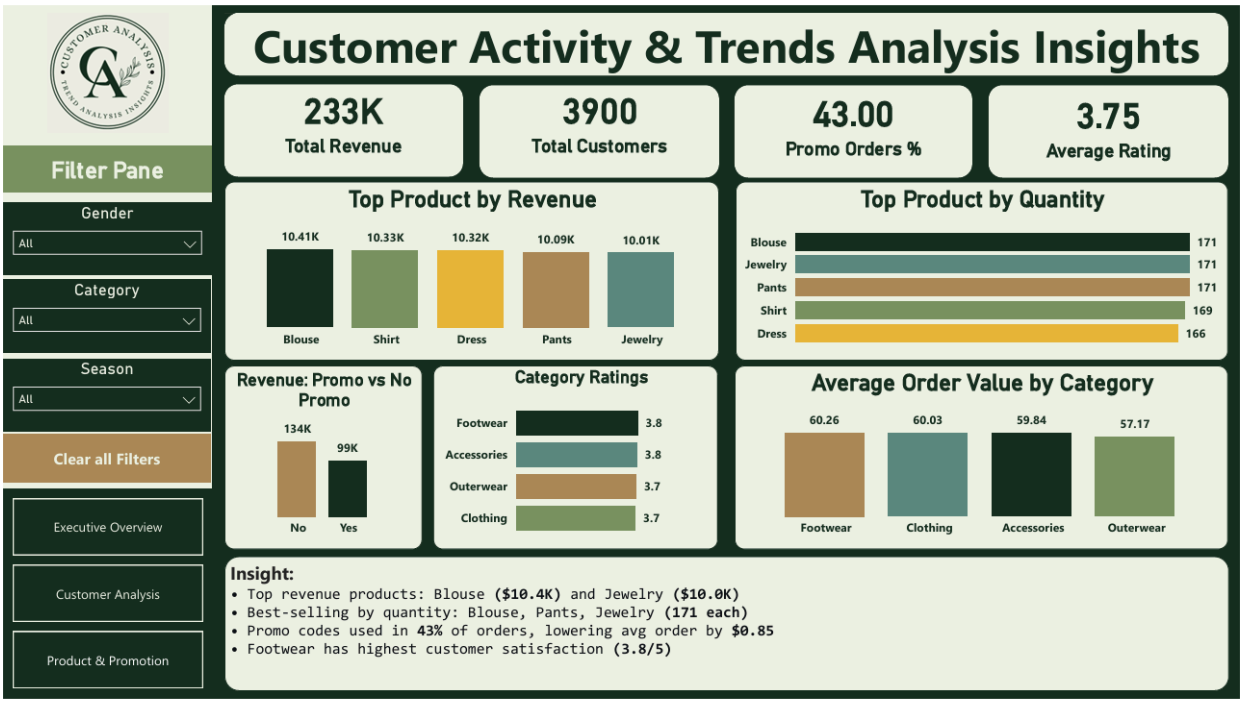
Customer Analysis

- Revenue by Customer Segment
- Customer Distribution
- Average Purchase by Age Group



Product and Promotion Analysis

- Top Products by Revenue
- Category Ratings
- Promotion Impact Analysis



7. Key Insights

The analysis produced several important insights:

- Clothing is the highest revenue generating product category.
- Loyal customers contribute the majority of total revenue.
- Middle-aged customers represent the highest spending customer segment.
- Fall season records the highest total sales compared to other seasons.
- Promotions and discounts have limited impact on increasing the average purchase amount.
- Footwear products receive the highest customer satisfaction ratings.

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8. Business Recommendations

Based on the insights generated from the analysis, several business recommendations can be made:

- Focus marketing campaigns on loyal customers to maximize retention.
- Increase promotion of high-performing product categories such as clothing.
- Develop targeted marketing strategies for high-spending age groups.
- Improve promotional strategies to increase their impact on sales.
- Increase marketing efforts during lower-performing seasons.

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9. Conclusion

This project demonstrates how data analysis can help businesses better understand customer behavior and purchasing patterns. By combining Python, SQL, and Power BI, the analysis provides valuable insights into customer segments, product performance, and seasonal sales trends.

The results of this project can support data-driven decision making and help businesses improve marketing strategies, customer engagement, and overall revenue growth.